Optimal Exiting for Liquidity Provision in Constant Function Market Makers

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Abstract

Providing liquidity to constant function market makers is often less profitable or favorable than simply holding assets, primarily because of impermanent loss. Using an optimal stopping approach, we show the existence of liquidity provision strategies that are profitable (excluding infrastructural fees) relative to holding, in which a liquidity provider (LP) exits the pool when the price ratio first hits certain thresholds. For the constant product pricing function, we derive closed-form solutions for the optimal thresholds. We demonstrate that pricing functions can be designed to maximize the expected time an optimally-acting LP exits the pool and backtest our strategy on Uniswap v2 data.

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1 Introduction

Decentralized exchanges (DEXs) are the leading category within decentralized finance (DeFi), representing 40% of the market share.[1] DEXs enable users to provide liquidity and trade directly on a peer-to-peer system, eliminating the need for intermediaries. Most DEXs utilize automated market makers (AMMs), enabling liquidity providers (LPs) and traders to interact with a *liquidity pool* instead of being directly matched with each other. Pricing is determined through predefined *pricing functions*. The vast majority of AMMs use the *constant function market maker (CFMM)* mechanism, where trading and liquidity provision are governed by the requirement to maintain a specific invariant of a bivariate function. A distinguishing feature of CFMMs is the introduction of *passive liquidity provision*: unlike traditional limit order books, where market makers actively manage liquidity through continuous submission of limit orders, users in a DEX can deposit liquidity into the pool without needing to constantly offer quotes or monitor the market.

However, liquidity provision in DEXs is often unprofitable, particularly in pools where at least one coin is risky, both in absolute terms and also relative to simply holding one's assets. One significant factor contributing to this unprofitability is the *impermanent loss* created by the CFMM pricing function, where an LP's deposited assets change in quantity and value due to arbitrage trades on the DEX (Capponi and Jia (2021)). Given the importance of liquidity pool depth on execution prices for traders and the functionality of DEXs as a whole, many market design solutions have been proposed to alleviate or eliminate the issue caused by impermanent loss and arbitrage risk. These include modifications to blockchain infrastructure (Milionis, Moallemi, and Roughgarden (2022b)), adaptive fees based on token pair volatility (Cao, Falk, and Tsoukalas (2024)), optimally designing the pricing function for LPs (Capponi and Chen (2024)), and allowing for novel forms of liquidity provision such as just-in-time liquidity (Capponi, Jia, and Zhu (2023)).

This paper examines the decision-making challenge faced by liquidity providers (LPs) in determining an optimal policy for exiting a liquidity pool, with the goal of balancing trading fee revenue against impermanent loss.[2] We consider a CFMM managing a pair of tokens, a risky and stable coin, and model the LP's decision problem as an optimal stopping problem. The LP selects an exit policy that maximizes their payoff relative to just holding their assets off chain. This payoff consists of a running reward which captures fee revenue earned from providing liquidity to the liquidity pool and a terminal penalty capturing the impermanent loss, the difference in value of assets deposited in a liquidity pool relative to holding them outside the pool due to traders swapping with the pool, incurred at the time of withdrawal. As the impermanent loss heavily depends on the risky coin's price at time of withdrawal compared to the price at time of deposit, a natural class of exit policies is given by first hitting times of the price process, i.e. the LP withdraws whenever the risky coin's price hits a threshold. In our model, given the price at the time when the LP supplied liquidity, the LP chooses an upper and lower price threshold to dictates their exit policy.

[1] For details, see `https://www.coingecko.com/research/publications/defi-categories-market-share`.

[2] Li et al. (2024) study a model of active LPs who can *continuously* rebalance between their personal wallets and liquidity pool, in contrast with our one-shot decision.

Since the downside risk of impermanent loss is bounded above, there is no lower price threshold at which it is more beneficial for an LP to withdraw liquidity than to remain invested in the pool; LPs can simply recover their bounded impermanent loss by earning fees from trades on the DEX. In contrast, the upside risk of impermanent loss is unbounded, and we demonstrate that for a broad class of pricing functions, there exists a unique upper price threshold that maximizes an LP's payoff relative to simply holding the assets. It is important to note that while tokens with higher volatility tend to be prone to higher impermanent loss, the primitive factor that governs impermanent loss is the difference in price relative to the price at deposit. While price changes in both directions result in impermanent loss, price increases are of greater concern as the potential losses can be unbounded. The assymetry of limited downside but unlimited upside resembles going long a put option or short a call option, revealing a similarity between liquidity provision on AMMs and traditional derivatives markets.[3]

Importantly, we show that a liquidity provider's expected payoff when using the optimal price threshold strategy is greater than the payoff resulting from holding the assets off-chain. This implies that, if one disregards infrastructural fees paid to the builders and validators involved in the block production process, also referred to as *gas fees*, liquidity provision can be profitable in expectation. When gas fees are low or negligible, as is the case on typical Layer 2 networks, our model provides an alternative strategy for LPs to mitigate losses: rather than altering the market design of DEXs, LPs can strategically time their exits from the pool to optimize their outcomes.

Our optimal stopping approach can be used to design pricing functions based on how long an optimally-behaving LP will keep their liquidity in the pool. Under this framework, a higher optimal upper price threshold, which corresponds to a longer expected time in the pool, is more desirable as it keeps an optimally-exiting LP in the pool for longer. Unlike existing approaches that emphasize welfare (Capponi and Jia (2021)) or trading efficiency (Goyal et al. (2023)), our approach focuses exclusively on value locked in the AMM pool.

We derive a formula for impermanent loss that depends solely on token prices and the curvature of the pricing function. This formula reflects the heuristic that greater curvature (or convexity) in the pricing function correspond to lower impermanent loss.[4] When market parameters, particularly the fee reward rate, are held constant, selecting a pricing function with a larger overall curvature increases the optimal upper price threshold. However, in practice, greater curvature also increases slippage for traders, which may reduce the overall trading volume on the AMM. To address this, we examine a parametric family of pricing functions where the "overall curvature" increases in the parameter value while the fee reward rate decreases in the parameter value. Under certain conditions on the parametric family and fee structure, we demonstrate that there exists a unique parameter value that maximizes the LP's optimal upper price threshold. Furthermore, this maximizer lies in the interior of the parameter space, indicating that the optimal AMM design based on our criterion achieves a balance between the benefits to traders and LPs.

[3]Milionis et al. (2022) provides another similarity, deriving an equivalence between the loss-versus-rebalancing metric and a bet on asset price volatility.

[4]See Capponi and Jia (2021) and Fabi and Prat (2023), for example.

Literature Review. Our study contributes to the literature on arbitrage risk facing liquidity providers in AMMs. Capponi and Jia (2021) show that arbitrage loss to passive LPs is inevitable in AMMs given the pricing function and transaction execution mechanisms of blockchains, which may cause liquidity freezes when price ratios are highly volatile. Their model is extended in Capponi, Jia, and Zhu (2023) to delineate conditions under which the active liquidity provision mechanism of *just-in-time liquidity* benefits passive LPs. Hasbrouck, Rivera, and Saleh (2022) demonstrate the importance of transaction fees in AMMs to attract liquidity and counteract arbitrage losses. Fabi and Prat (2023) study the economics of CFMMs, demonstrating that the interests of traders and LPs are at odds with each other.

Many studies within this stream of literature model time as continuous. Milionis et al. (2022) introduce and study the dynamics of *loss-versus-rebalancing* (LVR), an arbitrage loss metric when LPs can hedge their position via a rebalancing trading strategy. Milionis, Moallemi, and Roughgarden (2022b) examine LVR under Poisson block times and non-negligible transaction fees for arbitrageurs. Li et al. (2023) study a stochastic control problem for LPs that incorporates fee APR, impermanent loss, and gas fees. They find that LPs enter pools when APR is high and gas fees are low, and exit pools when the opposite holds.

Our work contributes to this strand of literature by taking an novel optimal stopping approach to liquidity provision. Noting that impermanent loss is highly depend on token prices at time of entry and exit, we examine the incentives behind liquidity provision when LPs can carefully choose when to exit the pool. Whereas previous works typically treat LPs as passive or active in some form (e.g. rebalancing, just-in-time liquidity), the LP in our model can be considered as "semi-active" due to their ability to time their exit.

We provide another method to mitigate losses for liquidity providers. Existing literature has largely focused on changes to blockchain infrastructure and the market design of DEXs. Milionis, Moallemi, and Roughgarden (2022b) suggest that faster intervals between the production of blocks reduces arbitrage profits. Canidio and Fritsch (2023) propose an AMM design with batch auctions that eliminates arbitrage profits in expectation. Capponi, Jia, and Zhu (2023) show that different fee tiers for active and passive LPs allow just-in-time liquidity to benefit passive LPs. Adams et al. (2024) propose auctions for "pool manager" rights and dynamic fees to achieve higher levels of liquidity on AMM pools. Relative to these works, we instead focus on what an LP can do themselves to mitigate arbitrage losses within the classical framework of a CFMM with fixed fees. Our model suggests that optimally-timed withdrawals from a liquidity pool resolve the impermanent loss issue and generate profits in expectation.

Some studies focus on liquidity provision strategies for AMMs that use the "concentrated liquidity" mechanism, such as Uniswap v3 (Adams et al. (2021)). Fan et al. (2023) use a framework based on neutral networks to solve a dynamic optimization problem for LPs, and Cartea, Drissi, and Monga (2024) formulate a stochastic control problem to derive optimal LP "spreads" on Uniswap v3. These strategies are tailored to sophisticated LPs, whereas our model and results are implementable by more passive LPs.

There is also a body of literature on designing the optimal pricing function given some criterion for evaluation. Capponi and Jia (2021) derive the socially optimal pricing function from a parametric family of pricing functions. Goyal et al. (2023) find optimal pricing functions given a distribution of price ratios based on a measure of trading efficiency. Milionis, Moallemi, and Roughgarden (2022a) take a Myersonian approach to designing AMMs based on an LP's beliefs on trading demand and token prices. Compared with the rest of the literature, we offer a novel criterion for pricing function optimization based on how long an optimally-exiting LP stays in a liquidity pool. Our model is applicable to various parametric families of pricing curves that satisfy certain properties.

2 Background

2.1 Decentralized Exchanges

Decentralized exchanges allow for the liquidity provision and trading of tokens on blockchains via smart contracts. DEXs consist of automated market makers: smart contracts that use *liquidity pools* and *pricing function* to algorithmically determine quotes for traders. In contrast to limit order books in which buyers and sellers are matched, liquidity providers and liquidity demanders interact with the liquidity pool rather than with each other. Many AMMs support exchanges between two tokens, but there exist a sizable number of AMMs that allow for exchanges between three or more tokens (e.g. Balancer and Curve).

LPs deposit tokens into the liquidity pool, with the relative quantities of each token determined by the pricing function that preserves the pre-existing marginal exchange rate, or *spot rate*, in the pool. In return, the LP receives some certificate demonstrating their stake in the pool based on the value of their deposit. In Uniswap v2, for example, LPs receive some amount of *pool tokens* that represents their proprotional stake in the pool, while in Uniswap v3, a single non-fungible token (NFT) tracks their liquidity position. LPs may choose to exit from the pool at any time, receiving their stake of the liquidity pool's value upon withdrawal.

Liquidity demanders, or traders, submit orders to exchange one token for another. The traded quantities must preserve an invariant determined by the pricing function. The quantities of valid trades implicitly determine relative prices for swapping. For most pricing functions, trading costs increase in the trade size and decrease in the depth of the liquidity pool. In additional to the price impact generated by the pricing function, traders also pay a transaction fee proportional to their trade size, with the proportional rate typically being pre-determined by the AMM. For example, Uniswap v2 sets a fixed 0.3% fee across all of its pools, while Uniswap v3 allows for multiple fee tiers, ranging from 0.01% to 1%. The trading fee either goes into the liquidity pool, directly to the LPs, or to the AMM. This incentivizes LPs to deposit and contributes to the upkeep of the AMM.

2.2 Constant Function Market Makers

The oldest type of AMM is the *constant function market maker (CFMM)*. To offer an example, we consider the *constant product* pricing function $F : \mathbb{R}_+^2 \to \mathbb{R}_+$ for a CFMM between tokens X and Y,

given by $F(x, y) = xy$, where x and y are the quantities of token X and Y, respectively. To maintain the CFMM's spot rate, LPs need to deposit equal values of both tokens. Suppose that there are 10 tokens X and 20 tokens Y already in the pool, with token X having twice the price of token Y. An LP can then deposit 2 tokens X and 4 tokens Y into the pool for a one-sixth stake in the pool. With the resulting pool reserves being $x = 12$ and $y = 24$, the CFMM invariant is $F(x, y) = 288$. A trader can then trade in 6 tokens X for 6 tokens Y, moving the reserves to $x' = 16$ and $y' = 18$, preserving the invariant level of 288 under F. Observe that the execution price is 1 while the fundamental price ratio is 1/2. If the LP were to withdraw after the trade, their one-sixth stake means that the LP receives back 8/3 tokens X and 3 tokens Y upon exit. A more formal treatment of CFMMs is found in the next section. Recent innovations in AMM design have introduced new types of AMMs. One such example is the *concentrated liquidity market maker (CLMM)*, which allows for LPs to choose the price range in which their liquidity can be used for trades, making CLMMs somewhat resemble traditional LOBs. For the remainder of this paper, we restrict our attention to CFMMs.

2.3 Impermanent Loss

One particular feature of CFMMs is *impermanent loss*: when changes in fundamental prices and arbitrage trades result in an LP's value in the liquidity pool being lower than an LP's value from simply holding the tokens. To see a simple example of this, suppose that initial price of token X is $p_X = 1$ and that token Y is the numeraire. An LP deposits 2 tokens X and 2 tokens Y into an AMM with the pricing function $F(x, y) = xy$ and no pre-existing liquidity. Now, suppose that p_X jumps to 4, and a arbitrageur trades in 2 tokens Y for 1 token X, bringing the pool reserves to $x = 1$ and $y = 4$. The value of the pool is $4 \times 1 + 1 \times 4 = 8$, while the value from holding the original quantity of assets, evaluated at the currenct token prices, is $4 \times 2 + 1 \times 2 = 10$. The difference $8 - 10 = -2$ is the impermanent loss. As p_X further increases, the impermanent loss worsens, and LPs incur impermanent loss when p_X decreases as well. However, if an LP withdraws from the pool when the price of token X is the same as that at the time of deposit, then impermanent loss is completely mitigated due to reverse trades by arbitrageurs. Our paper takes a particular focus on this aspect of CFMMs, balancing loss mitigation with revenue from transaction fees.

3 Model

Asset Price Dynamics. Time is continuous on $\mathbb{R}_+$. Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in\mathbb{R}_+}, \mathbb{P})$ be a filtered probability space and $\{W_t\}$ be an $\{\mathcal{F}_t\}$-adapted Wiener process. There is a risky coin X with fundamental price P_t at time t and a stable coin Y with price 1 at all times. The risky coin's price process $\{P_t\}$ follows a geometric Brownian motion governed by the stochastic differential equation

$$dP_t = \mu P_t \, dt + \sigma P_t \, dW_t$$

where $\mu \in \mathbb{R}$ is the drift parameter and $\sigma > 0$ is the volatility parameter. The risk-free rate is $r > 0$.

Constant Function Market Maker. There is a CFMM for exchanging the risky and stable coins with fee tier f. We work with the class of pricing functions that satisfy Assumption 1 below.[5]

Assumption 1. *The CFMM's pricing function $F : \mathbb{R}_+^2 \to \mathbb{R}_+$ satisfies the following properties:*

(i) $F_x, F_y > 0$ *and* $F_y^2 F_{xx} - 2F_x F_y F_{xy} + F_x^2 F_{yy} < 0$*;*

(ii) F *is k-homogeneous, i.e. there exists $k > 0$ such that $F(cx, cy) = c^k \cdot F(x, y)$ for all $c > 0$;*

(iii) $\lim_{x\to 0} F_x/F_y = \infty$, $\lim_{x\to\infty} F_x/F_y = 0$, $\lim_{y\to 0} F_x/F_y = 0$, *and* $\lim_{y\to\infty} F_x/F_y = \infty$.

Liquidity provision and trading on the CFMM works as follows. Suppose that at time t, there are X_t risky coins and Y_t stable coins. A liquidity provider who wishes to deposit d_X risky coins and d_Y stable coins at time t must have d_X and d_Y preserve the liquidity pool's *spot rate*, which we assume is aligned with the fundamental price due to the presence of arbitrageurs:

$$\frac{F_x(X_t + d_X, Y_t + d_Y)}{F_y(X_t + d_X, Y_t + d_Y)} = P_t.$$

Let $\mathcal{I}_t = F(X_t, Y_t)$. A trader who wishes to trade q_X risky coins for q_Y stable coins at time t must have q_X and q_Y preserve the liquidity pool's *invariant level* $\mathcal{I}_t$ prior to the trade:

$$F(X_t + q_X, Y_t + q_Y) = \mathcal{I}_t.$$

Arbitrageurs, Pool Value Function, and Impermanent Loss. We assume that a population of arbitrageurs continuously monitors the CFMM's liquidity pool to align the CFMM's marginal price, i.e. F_x/F_y, with the fundamental price of the risky coin. This assumption captures the fact that due to priority fee bidding, LPs almost always have their transactions ordered after those of arbitrageurs.

Milionis et al. (2022) show that given an invariant level $\mathcal{I}$, the value of pool reserves under the presence of arbitrageurs when the price is P, denoted by $V(P; \mathcal{I})$ or $V(P)$ in short, solves the *pool value problem* given by

$$\begin{aligned} V(P; \mathcal{I}) = \min_{(x,y)\in\mathbb{R}_+^2} & (Px + y) \\ \text{subject to } & F(x, y) = \mathcal{I}. \end{aligned}$$

We show in the appendix that $V(P)$ is well-defined when F satisfies Assumption 1. Capponi and Jia (2021) show that arbitrageurs solve a similar problem, framed in terms of the CFMM's *trading function* and incorporating a transaction fee.

Impermanent loss can be concisely characterized in terms of the pool value function. Suppose that a liquidity provider deposits x_0 risky coins and y_0 stable coins at time 0. Let $I_0 = F(x_0, y_0)$.

[5] Assumption 1 results in a slightly broader class of pricing functions than the class considered in Capponi and Jia (2021): the "convexity" property ($F_{xx} < 0$, $F_{yy} < 0$ and $F_{xy} > 0$) in Capponi and Jia (2021) is sufficient for $F_y^2 F_{xx} - 2F_x F_y F_{xy} + F_x^2 F_{yy} \leq 0$.

In the appendix, we show that this LP's share of the total pool value is equivalent to solving the pool value problem at invariant level I_0. In other words, let the total pool reserves at time t, under the presence of arbitrageurs, consist of X_t risky coins and Y_t stable coins. Let $\mathcal{I}_t = F(X_t, Y_t)$ and w_t be the LP's share of the pool at time t. Then

$$w_t \cdot V(P_t; \mathcal{I}_t) = V(P_t; I_0)$$

Note that this LP's share of the pool value only depends on the price process and the *initial* invariant level of the deposit. Had the LP held the assets, their value at time t, denoted by V_h, would be

$$V_h(P_t) = P_t x_0 + y_0.$$

The *per-unit impermanent loss* of the LP, denoted by L, is the difference in value between holding x_0 risky coins and y_0 stable coins versus providing liquidity to the CFMM, normalized by x_0, i.e.

$$L(P_t) = \frac{V(P_t; I_0) - V_h(P_t)}{x_0}.$$

Since $V_h(P_t)$ is the value of a suboptimal solution to a minimization problem whose optimal value is given by $V(P_t; I_0)$, it follows that $L(P_t) \leq 0$ for all $P_t \in \mathbb{R}_+$.

Noise Traders and Trading Fees. There is a population of noise traders such that the combined instantaneous trading volume (of both the arbitrageurs and noise traders) is of the form $\nu \cdot x^*(P_0; \mathcal{I}_t)$, where $x^*(P_0; \mathcal{I}_t)$ solves the pool value problem at P_0 ant $\mathcal{I}_t$. The increasing trading intensity with respect to pool depth reflects the increased order flow volume attracted by a deeper market with better execution prices, and the specific functional form allows an individual liquidity provider's instantaneous *fee reward rate*, the pro-rata share of fees denoted by $\phi \in \mathbb{R}_+$, to be constant.[6]

The Liquidity Provider's Problem. We consider the perspective of a single liquidity provider in a CFMM, focusing on the management of a liquidity position in a one-shot setting, and do not consider subsequent positions that the LP may take. Our modeling assumptions result in the LP's problem being agnostic to the behavior of and strategic interaction between other LPs in the pool. At time $t = 0$, the LP deposits x_0 risky coins and y_0 stable coins in the liquidity pool of the CFMM. While the LP's assets are locked in the pool, the LP earns an per-unit instantaneous fee reward rate of $\phi \in \mathbb{R}_+$. Upon withdrawal at a stopping time T (with respect to $\mathcal{F}$), the LP incurs a per-unit impermanent loss of $L(P_T)$. Discounting by rate $r > 0$, the LP's expected payoff relative to holding the same quantity of assets is thus

$$\mathbb{E}\left[\int_0^T e^{-rt}\phi\, dt + e^{-rT} \cdot L(P_T)\right].$$

[6]The assumption that noise trading volume increases in pool size can be microfounded via a noise trader that has a private value of the risky coin (see Capponi, Jia, and Zhu (2023), for instance).

Restricting the action set of the LP to a family of stopping times $\mathcal{T}$, the LP's problem is to choose the stopping time that maximizes their payoff relative to holding, i.e.

$$\sup_{T \in \mathcal{T}} \mathbb{E}\left[\int_0^T e^{-rt}\phi\, dt + e^{-rT} \cdot L(P_T)\right].$$

The family of stopping times we consider is the one generated by two-sided hitting times. For $P_L < P_0$, let $T(P_L) = \inf\{t \in \mathbb{R}_+ : P_t \leq P_L\}$ and for $P_U > P_0$, let $T(P_U) = \inf\{t \in \mathbb{R}_+ : P_t \geq P_U\}$. Let $T(P_L, P_U) = \min\{T(P_L), T(P_U)\}$; then we can express $\mathcal{T}$ as

$$\mathcal{T} = \{T(P_L, P_U) : P_L < P_0,\ P_U > P_0\}.$$

Discussion of Assumptions and AMM Loss Metrics. Although blockchain transactions are settled in discrete blocks, we assume that time is continuous (similar to Milionis et al. (2022) and Li et al. (2023), for example), allowing us to use stochastic control theory and HJB equations to solve the LP's problem. Whereas several studies micro-found trading behavior on DEXs, we instead use a reduced-form model of trading given by an intensity that increases with respect to the invariant level. The trading intensity scales in such a way that any LP's fee rewards from providing liquidity is constant regardless of the LP's stake in the pool; the constant rate ϕ is critical to the tractability of the HJB equation and independence of the LP's problem on other LPs in the liquidity pool.

Impermanent loss and loss-versus-relabancing (LVR) are two related but different measures of value lost to arbitrageurs on CFMMs. Milionis et al. (2022) describe impermanent loss as "loss-versus-holding" and show that it can be decomposed into the sum of two components: LVR and market risk, where the market risk component can be either positive or negative. LVR thus measures the risk of providing liquidity that cannot be hedged away by a rebalancing trading strategy, while market risk is incorporated into the impermanent loss. For more passive or semi-active LPs, impermanent loss may be more applicable since these types of LPs could be unable or not well-capitalized enough to implement the rebalancing strategy. Additionally, LVR is path-dependent and cannot be used as the "terminal reward" component in a standard stochastic control setting.

4 Optimal Exiting Policies

We reparameterize the LP's problem in terms of log-prices. Let $p_t = \log P_t/P_0$ denote the log-(relative) price. Then the log-price process follows the stochastic differential equation given by

$$dp_t = \tilde{\mu}\, dt + \sigma\, dW_t$$

where $\tilde{\mu} = \mu - \sigma^2/2$. Let ℓ denote the per-unit impermanent loss function in terms of log-prices with the relation between the two given by

$$\ell(p_t) = L(P_0 \exp p_t).$$

As the fee reward rate does not depend on price, the reparameterized problem is given by

$$\sup_{T \in \mathcal{T}} \mathbb{E}\left[\int_0^T e^{-rt}\phi\, dt + e^{-rT} \cdot \ell(p_T)\right].$$

The reparameterized family of stopping times is now

$$\mathcal{T} = \{T(p_L, p_U) : p_L < 0,\, p_U > 0\}$$

where $T(p_L) = \inf\{t \in \mathbb{R}_+ : p_t \leq p_L\}$ for $p_L < 0$ and $T(p_U) = \inf\{t \in \mathbb{R}_+ : p_t \geq p_U\}$ for $p_U > 0$. We can write the problem as

$$\sup_{(p_L, p_U): p_L < 0, p_U > 0} \mathbb{E}\left[\int_0^T e^{-rt}\phi\, dt + e^{-rT} \cdot \ell(p_T)\right].$$

In other words, we are interested in the optimal exit policy that can be characterized by an log-price interval (p_L, p_U) where $p_L < 0$ and $p_U > 0$ such that the LP exits the pool when the log-price is first outside this interval.

Theorem 4.1. *There exists a unique optimal solution (p_L^*, p_U^*) to the LP's problem in which:*

- $p_L^* = -\infty$*;*
- $p_U^* \in (0, \infty)$ *if* $r \leq \mu$ *and* $p_U^* = \infty$ *otherwise;*
- $v(0; p_L^*, p_U^*) > 0$*, i.e. expected payoff is positive.*

This means that there is no lower price threshold at which withdrawing yields a higher payoff in expectation relative to holding. As impermanent loss is bounded when the price decreases, given the same fee reward rate ϕ, it is better for LPs to stay in the pool and recuperate their impermanent loss via fees. Thus, the downside risk of providing liquidity to a CFMM is bounded and can be covered by incentives for liquidity provision. However, upside impermanent loss is unbounded, which significantly affects how the optimal upper price threshold is determined.

On the other hand, impermanent loss is unbounded as price tends to infinity. For the LP, there is now a tradeoff between worsening impermanent loss as the price increases and the inability to earn fees upon withdrawal from the pool. The above theorem shows that for a CFMM pricing function that satisfies Assumption 1, there exists a unique policy that maximizes expected payoff relative to holding among all upper price threshold policies. Moreover, the fact that $v(0) > 0$ implies that the optimal policy yields a positive expected payoff relative to holding, net of infrastructural fees, giving rise to a profitable liquidity provision strategy when network fees are negligible.

In the case where μ is not sufficiently large relative to r, then the optimal upper price threshold is also infinite. The intuition is that for fixed stopping times T, smaller values of μ correspond to longer expected times in the pool, i.e. larger $\mathbb{E}[T]$. This means that there are more fees accumulated and a smaller discounted impermanent loss incurred, with upon exiting the pool, with the latter

force outweighing the former, relative to a larger value of μ which corresponds to a shorter expected time in the pool.

Comparative Statics with Respect to Price and Trading Dynamics. Having established the existence and uniqueness of an optimal upper price threshold for a class of pricing functions, we now examine how this threshold varies as market parameters, i.e. the price process and fee reward rate change. Besides the market parameters, the other main determinant of the optimal threshold is the pricing function, which we take a closer look at in the following section. The following result characterizes the comparative statics.

Theorem 4.2. *If r is sufficiently small,*[7] *then the optimal upper price threshold p_U^* increases in ϕ and decreases in μ and σ.*

These results are fairly intuitive. A larger value of μ, meaning a more positive drift of the price process, drives p_U^* down as the upside risk increases, raising the chance for potentially unbounded losses. Higher volatility increases price uncertainty, leading to a similar increase in the upside risk. This finding is in line with other results in the literature, particularly that LVR is proxied by volatility and the quadratic variation of the price process. Thus, upward price movement and volatility disincentivize liquidity provision due to a larger expected impermanent loss. An increase in the fee reward rate, on the other hand, incentivizes liquidity provision as LPs will want to stay longer in pool and earn the extra fees, holding all other parameters fixed.

Example: Constant Product Function We derive the impermanent loss function and resulting optimal upper price threshold for the constant product function as an example. The constant product function, used on Uniswap v2 and on many other DEXs, is given by $F(x, y) = xy$. Solving the pool value problem for invariant level $\mathcal{I}$ yields optimal reserves and a pool value of

$$(x^*(P), y^*(P)) = \left(\mathcal{I}^{1/2}P^{-1/2}, \mathcal{I}^{1/2}P^{1/2}\right),$$

$$V(P) = 2\mathcal{I}^{1/2}P^{1/2}.$$

For the constant product function, the LP's initial deposit of x_0 risky coins and $y_)$ stable coins must satisfy $y_0/x_0 = P_0$ and $x_0y_0 = \mathcal{I}_0$, so we have

$$V_h(P) = Px_0 + y_0(P + P_0)x_0,$$

$$V(P) = 2(x_0y_0)^{1/2}P^{1/2} = 2P_0^{1/2}P^{1/2}x_0.$$

Thus, the impermanent loss function is given by

$$L(P) = \frac{V(P) - V_h(P)}{x_0} = 2P_0^{1/2}P^{1/2} - (P_0 + P).$$

[7] Specifically, we require $r \leq \mu$ for p_U^* to be finite.

Note that the impermanent loss under the constant product function is twice the difference between the geometric and arithmetic means. Let $p = \log P/P_0$. Then the impermanent loss function in log-prices is

$$\ell(p) = P_0 \left(2 \exp \frac{p}{2} - \exp p - 1\right).$$

Solving the optimal stopping problem results in an optimal upper price threshold p_U^* of

$$p_U^* = 2 \log \frac{1 - 2\xi^+ + \sqrt{1 + \frac{4}{P_0}\xi^+(1-\xi^+)\frac{\phi}{r}}}{2(1-\xi^+)}$$

where ξ^+ is the positive solution to $r = \tilde{\mu}\xi + \frac{1}{2}\sigma^2\xi^2$. We can then verify that the comparative statics results of Theorem 4.2 hold for the closed-form solution given above.

5 Optimal Pricing Functions

Comparative Statics with Respect to the Pricing Function. Having analyzed the comparative statics of the optimal upper price threshold with respect to market parameters, we now explore how the shape of the pricing function itself affects the optimal threshold. The pricing function affects the LP's optimal stopping problem through the impermanent loss function in the terminal reward (or penalty in this case), which we find is closely related to the curvature of the pricing function. Our definition of the curvature function given a pricing function, price level, and invariant level, is provided below. This is the standard notion of curvature for the level set of a parametric curve in two dimensions, in this case given by $F(x, y) = \mathcal{I}$. In this context, the curve is parameterized by the token price P and evaluated at the no-arbitrage reserve quantities at P.[8]

Definition. *Let F be some pricing function. The* <u>*curvature function*</u> *of F at price P and invariant level $\mathcal{I}$, denoted $\kappa(P;\mathcal{I})$, is*

$$\kappa(P;\mathcal{I}) = \left.\frac{|F_y^2 F_{xx} - 2F_x F_y F_{xy} + F_x^2 F_{yy}|}{(F_x^2 + F_y^2)^{3/2}}\right|_{(x^*(P;\mathcal{I}), y^*(P;\mathcal{I}))}$$

where $(x^*(P;\mathcal{I}), y^*(P;\mathcal{I}))$ solves the pool value problem at price P and invariant level $\mathcal{I}$.

Proposition 5.1. *If a LP deposits x_0 risky coins and y_0 stable coins at $t = 0$, then the LP's per-unit impermanent loss function L at price P is*

$$L(P) = \frac{1}{x_0} \int_{P_0}^{P} \frac{\pi - P}{\kappa(\pi; I_0)} \left(\frac{1}{1+\pi^2}\right)^{3/2} d\pi$$

where $I_0 = F(x_0, y_0)$.

[8] The curvature function is connected to the notion of "slippage" in Fabi and Prat (2023): specifically, $\kappa(P)$ is equal to the slippage evaluated at reserves $(x^*(P), y^*(P))$ scaled by $(1 + P^2)^{-3/2}$.

With this notion of curvature, we can express the impermanent loss as a function of price and curvature. The formula in Proposition 5.1 reveals that impermanent loss can be expressed as a weighted integral of the reciprocal of the curvature function. The "weights" in the integral are larger for price levels closer to P_0 and increase in P. For larger curvatures, the amount contributed to the integral is smaller, leading to the heuristic of pricing function's with larger curvatures having smaller impermanent losses. We formalize a notion of comparing the curvature of two pricing functions in the definition below.

Definition. *Let $F^{(1)}$ and $F^{(2)}$ be pricing functions with curvature functions κ_1 and κ_2 at invariant levels $\mathcal{I}^{(1)}$ and $\mathcal{I}^{(2)}$, respectively. We say that $F^{(1)}$ <u>curvature-dominates</u> $F^{(2)}$ in the price range $[\underline{P}, \overline{P}]$ if $\kappa_1(P; \mathcal{I}^{(1)}) \geq \kappa_2(P; \mathcal{I}^{(2)})$ for all $P \in [\underline{P}, \overline{P}]$.*

As the curvature of a pricing function can vary significantly depending on the fundamental price and invariant level, to compare the curvatures of two pricing functions, we need to fix a price range and invariant level for comparison. When the fundamental price at time of deposit is within this price range, i.e. $P_0 \in [\underline{P}, \overline{P}]$, an immediate corollary is that if pricing function F_1 curvature-dominates F_2 on $[\underline{P}, \overline{P}]$, then the impermanent loss function of F_1 is less than that of F_2 on $[\underline{P}, \overline{P}]$. We would thus expect that if the fee reward rate ϕ remains the same and $F^{(1)}$ curvature dominates $F^{(2)}$ for all prices larger than P_0, then the optimal upper price threshold increases under a pricing function with larger curvature.

Proposition 5.2. *Suppose that a LP deposits x_0 risky coins and y_0 stable coins at $t = 0$. If $F^{(1)}$ and $F^{(2)}$ are pricing functions such that $F^{(1)}$ at invariant level $I_0^{(1)} := F^{(1)}(x_0, y_0)$ curvature-dominates $F^{(2)}$ at invariant level $I_0^{(2)} := F^{(2)}(x_0, y_0)$ in the price range $[P_0, \infty)$, then the optimal upper price threshold p_U^* in a CFMM with pricing function $F^{(1)}$ is greater than that in a CFMM with pricing function $F^{(2)}$.*

This result has implications for the design of AMMs. Assuming that LPs exit the pool optimally among upper price threshold policies, a larger price threshold implies that the expected time LPs spend with their assets locked in the liquidity pool is longer, given that market parameters are held constant. Indeed, under the same law of motion for the fundamental price process, the expected first hitting time for a price level increases in the price level itself. Keeping LPs in the pool for a longer time retains liquidity, which is beneficial for the AMM. Proposition 5.2 introduces a new paradigm for designing and evaluating pricing function by analyzing how long an optimally-behaving semi-active LP keeps their liquidity in a pool using that pricing function. It is important to note that for the above result, we fixed the fee reward rate ϕ, but in practice, higher curvature pricing functions generate more slippage, which may in turn affect the overall trading volume. We proceed to consider a setting in which the curvature also affects the fee reward rate.

Optimal Pricing Function from a Parametric Family. Fix P_0 and x_0, the amount of risky coins initially deposited by the LP. Let $\{F_\theta\}_{\theta \in \Theta}$ be a parametric family of pricing functions with parameter space $\Theta = [\underline{\theta}, \overline{\theta}] \subseteq \mathbb{R}_+$ where θ is a parameter capturing the curvature of the pricing

function such that higher values of θ correspond to pricing functions with larger curvatures. Specifically, if we have $\theta_1 > \theta_2$, then F_{θ_1} curvature-dominates F_{θ_2} on $[P_0, \infty)$, i.e. $\kappa_{\theta_1}(P) < \kappa_{\theta_2}(P)$ for all $P \in [P_0, \infty)$. The invariant level for the LP's position at θ, denoted $I_0(\theta)$, is the value of k that solves $x^*_\theta(P_0; k) = x_0$. To capture lower trading volumes induced by a pricing function with a larger curvature, we assume that the fee reward rate $\phi(\theta)$ is decreasing in θ. Our goal now is to find the value of $\theta \in \Theta$ that yields the highest optimal upper price threshold $p^*_U(\theta)$ for an LP who deposits x_0 risky coins at price P_0.

Theorem 5.3. *If $1/\kappa_\theta(P)$ is decreasing and convex in θ, and $\phi(\theta)$ is decreasing and concave in θ for $\theta \in [\underline{\theta}, \overline{\theta}]$ with $\phi(\underline{\theta}) < \infty$ and $\phi(\overline{\theta}) = 0$, then there exists a unique $\theta^* \in (\underline{\theta}, \overline{\theta})$ that maximizes the optimal upper threshold $p^*_U(\theta)$.*

The additional assumptions in Theorem 5.3 provide sufficient conditions for which there exists a unique maximizer in the *interior* of Θ. This choice of parameter, denoted by θ^*, maximizes the upper price threshold that an optimally-behaving LP uses, and thus the expected time an LP stays in the pool, out of the parametric family of pricing functions. The pricing function corresponding to parameter θ^* balances the competing forces of impermanent loss and fee revenue. This result is similar to Capponi and Jia (2021)'s derivation of the optimal pricing function in their model, where the optimal choice of curvature is in the interior of the parameter space, balancing both arbitrage loss and trading volume. Thus, desining pricing functions based on our optimal stopping approach is thus able to yield results related to those generated from other models of liquidity provision.

Example: Constant Elasticity Functions An example of a parametric family of pricing functions whose curvature for a fixed x_0 is increasing in the value of the parameter is the family of *constant elasticity* pricing functions $\{F_\theta\}_{\theta \in [1,\infty)}$ parameterized by $\theta \in [1, \infty)$ where

$$F_\theta(x, y) = (x^{-\theta} + y^{-\theta})^{-1/\theta}.$$

Observe that as $\theta \to \infty$, F_θ converges to a Leontief function. Solving the pool value problem for F_θ yields arbitrageur-optimal reserves of

$$x_\theta(P; \mathcal{I}) = \frac{\mathcal{I}(1 + P^{\theta/(1+\theta)})^{1/\theta}}{P^{1/(1+\theta)}}$$

$$y_\theta(P; \mathcal{I}) = \mathcal{I}(1 + P^{\theta/(1+\theta)})^{1/\theta}$$

and a curvature function of

$$\kappa_\theta(P) = \frac{1+\theta}{\mathcal{I}} \cdot (1 + P^{\theta/(1+\theta)})^{(\theta-1)/\theta} \left(\frac{P^{(4+2\theta)/(3+3\theta)}}{1 + P^2} \right)^{3/2}.$$

For a fixed $x_0 = x(P;\theta)$ and P_0, we have that

$$\kappa(P;\theta) = \frac{1+\theta}{x_0} \cdot \frac{P_0^{1/(1+\theta)}}{(1+P_0^{\theta/(1+\theta)})^{1/\theta}} \cdot (1+P^{\theta/(1+\theta)})^{(\theta-1)/\theta} \left(\frac{P^{(4+2\theta)/(3+3\theta)}}{1+P^2} \right)^{3/2}.$$

Standard calculus shows that $\kappa(P;\theta)$ is increasing in θ and $1/\kappa(P;\theta)$ is convex in θ for $\theta \in [1,\infty)$ at all $P \in \mathbb{R}_+$, so this parametric family satisfies the assumptions for Proposition 5.2 and Theorem 5.3. In the setting of the former result, when ϕ is constant in θ, choosing a higher value of θ results in a higher optimal upper threshold. However as slippages increase in θ as well, trading demand may not remain the same. For an appropriate trading demand schedule $\phi(\theta)$ that also satisfies Theorem 5.3's assumptions, there then exists an optimal choice of θ in the interior of the parameter space $[1,\infty)$.

6 Empirical Analysis

Data Collection. We collect pool- and transaction-level data from Uniswap v2 liquidity pools between 1 February 2022 and 1 August 2024. We focus on the top three pools consisting of Wrapped Ether (WETH) with stablecoins pegged to the US dollar, i.e. the WETH–USDT, USDC–WETH, and DAI–WETH pools. Using Amberdata, we obtain pool reserves at hourly frequencies and the token in/out quantities from all "swap" events occurring on the pool, which signify that a trade has occurred. We collect ETH–USD price data from centralized exchanges, also from Amberdata. For the discount rate, we obtain US T-Bill rates from the US Department of the Treasury.

Parameter Calibration. We proceed to calibrate the model parameters μ, σ, ϕ, and r.

- We first estimate the GBM parameters μ and σ for the price process using sample statistics of CEX price data. Observing daily log-returns of $\rho_1, \ldots, \rho_T$ in the CEX price time series during the "training" period, let

$$\bar{\rho} = T^{-1} \sum_{t=1}^{T} \rho_t.$$

  then the estimated GBM parameters are

$$\hat{\sigma} = \sqrt{\frac{\sum_{t=1}^{T} (\rho_t - \bar{\rho})^2}{T-1}},$$

$$\hat{\mu} = \bar{\rho} + \hat{\sigma}^2$$

- To estimate the fee reward rate ϕ, we first construct a time series of daily pro-rata fee revenue $\{f_t\}$ by taking the total daily volume (in stablecoin), multiply it by the fee tier of 0.3%, and divide by one plus the median amount of WETH reserves in that day. This corresponds to the

fee revenue earned by a hypothetical LP who deposits one unit of WETH and the appropriate quantity of stablecoins. We assume that adding this small amount of hypothetical liquidity to the pool does not impact observed real-world trading on the pool, which is realistic especially for large pools like WETH–USDT, USDC–WETH, and DAI–WETH. We then estimate ϕ as the mean parameter of a GARCH-M model fitted to $\{f_t\}$.

- We take the one-year US T-Bill rate, normalized to a daily frequency, as r.[9]

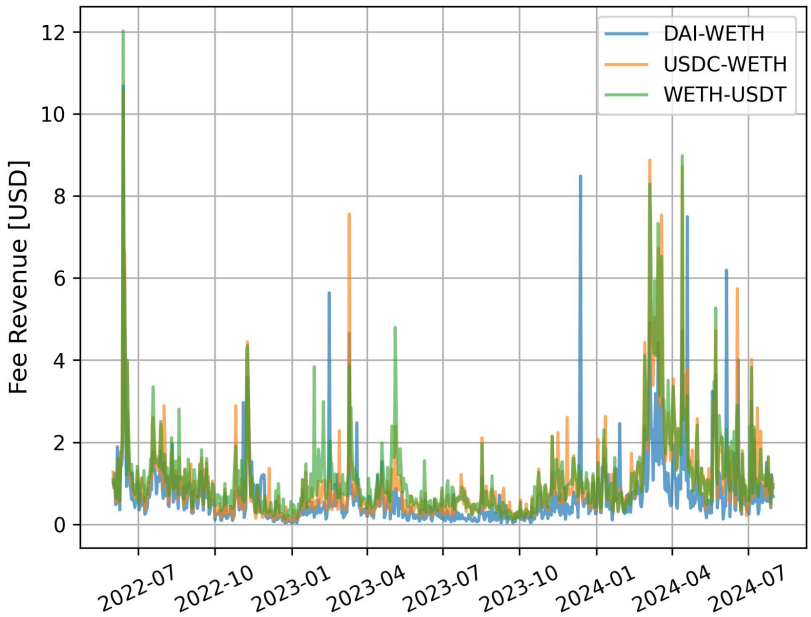

Figure 1: Time Series of Fee Revenue (USD) for Selected Pools

	DAI–WETH	USDC–WETH	WETH–USDT
N	790	790	790
Mean	0.6966	1.0130	1.1371
S.D.	0.8822	1.1123	1.0973
25%	0.2419	0.4171	0.5340
50%	0.4586	0.7042	0.8576
75%	0.7997	1.1154	1.3070

Table 1: Fee Revenue (USD) Summary Statistics for Selected Pools

Figure 1 and Table 1 plot the time series and display summary statistics for the fee revenue of an LP who hypothetically provides one unit of WETH to each pool. Based off the time series, fee

[9]See `https://home.treasury.gov`.

revenue generally fluctuates around a mean level with occasional periods of high volatility, which suggests that our constant fee reward rate assumption is reasonable.

Backtesting Procedure. Since the LP's problem maximizes expected payoff where the expectation is taken with respect to the price process, and due to the changes in calibrated parameters, we backtest the exit policy implied by our model with *bootstrapped data* and *adaptive updating* of the optimal exit thresholds.

For a designated start date, we go back $T_{train} = 90$ days from the start date for the training data and use data from the start date to the end of the sample period as the test set. For $t \in [T_{test}]$, let (ρ_t, f_t) be the observed log-returns and fee revenue at day t, respectively. Since we model the price process as a GBM, log-returns are assumed to be independent of each other, while fee revenue may be correlated with log-returns. To account for this, when resampling to obtain a simulated dataset of length T from the test data, we sample indices $\{i_1, \ldots, i_T\}$ independently with replacement from $[T_{test}]$, and 'couple" the observed fee revenue associated with the observed log-return by taking $((\rho_{i_1}, f_{i_1}), \ldots, (\rho_{i_T}, f_{i_T}))$ as our simulated data. We set $T = c \cdot T_{test}$ where $c = 2$.

To implement the exit policy with adaptive updating of the optimal thresholds, we construct two time series. The first is the simulated price $\{P_t\}_{t\in[T]}$ where $\tilde{P}_{t+1} = P_t \exp \rho_{i_t}$ and $\tilde{P}_0$ is the ETH price on the last training set day. The second is the optimal upper price threshold $\{(p_U^*)_t\}_{t\in[T]}$ for exiting the pool where $(p_U^*)_t$ is obtained by appending $(\rho_{i_1}, f_{i_1}), \ldots, (\rho_{i_t}, f_{i_t}))$, the first t days of simulated data, to the training data, then using the combined data to calibrate and solve the model for the optimal threshold. The exit time τ, if it exists, is the first time that the simulated price on a given day is higher than the optimal threshold computed at that same day, i.e.

$$\tau = \inf\{t \in [T] : P_t > (p_U^*)_t\}.$$

If $P_t < (p_U^*)_t$ for all $t \in [T]$, then the LP does not exit for the entire duration of the simulation, and we let $\tau = T$. We compute the LP's payoff is as

$$v^{strategy} = \sum_{t=1}^{\tau} e^{-rt} f_t + e^{-r\tau} \cdot L(P_\tau).$$

Note that if the LP does not exit during the simulated period, we use the end price of the simulated data, rather than that implied by the optimal threshold, to evaluate the impermanent loss.

For each simulated dataset, we also compute the LP's payoff had they used a fixed price P as their upper price threshold across a grid of price $P > P_0$. Letting $\tau(P) = \inf\{t \in [T] : P_t > P\}$ if $\tau(P) = T$ if $P_t \leq P$ for all $t \in [T]$, we define

$$v(P) = \sum_{t=1}^{\tau(P)} e^{-rt} f_t + e^{-r\tau(P)} \cdot L(P).$$

Results. We implement the backtesting procedure for a series of start dates, running $N = 1000$ simulations per start date. For simulation $i \in [N]$, let τ_i be the exit time and $v_i^{strategy}$ be the LP's payoff. Let $v_i(P)$ be the payoffs evaluated for exit policies based on a fixed threshold P. We keep track of the average payoffs $\bar{v}^{strategy}$ and $\bar{v}(P)$, as well as the average exit price threshold $\bar{P}_\tau$:

$$\bar{v}^{strategy} = N^{-1} \sum_{i=1}^{N} v_i^{strategy},$$

$$\bar{v}(P) = N^{-1} \sum_{i=1}^{N} v_i(P),$$

$$\bar{P}_\tau = N^{-1} \sum_{i=1}^{N} P_{\tau_i}.$$

We evaluate our strategy by comparing the strategy's average payoff to the maximum possible average payoff based on a fixed threshold P. We also compare the strategy's average exiting price to the argument of the maximum possible average payoff based on a fixed threshold P. Specifically, we look at the metrics $\mathsf{PctErrPayoff}$ and $\mathsf{PctErrPrice}$, given by:

$$\mathsf{PctErrPayoff} = \frac{\max_{P>P_0} \bar{v}(P) - \bar{v}^{strategy}}{\max_{P>P_0} \bar{v}(P)} \times 100,$$

$$\mathsf{PctErrPrice} = \frac{\arg\max_{P>P_0} \bar{v}(P) - \bar{P}_\tau}{\arg\max_{P>P_0} \bar{v}(P)} \times 100.$$

	DAI–WETH	USDC–WETH	WETH–USDT
Mean Exiting Time (Days)	414.19	484.14	511.49
Mean Initial Price (USD)	1491.67	1491.67	1491.67
Mean Optimal Exiting Price (USD)	2377.67	2847.67	3053.67
Mean Optimal Payoff (USD)	139.68	280.26	345.21
Mean Price Error	13.65%	11.95%	14.92%
Mean Payoff Error	16.73%	11.94%	11.57%

Table 2: Backtesting Results for Selected Pools

Our start dates are the 1st and 16th of every month from August to December 2022, inclusive. The summary statistics of average percent errors in the price thresholds and payoffs from optimality are reported in the Table 2, rounded to the nearest percent. The average errors are around 11–16% for both the price threshold and payoffs, with positive average optimal payoffs, as our theoretical results would suggest. We can also see how higher daily fee revenue (c.f. Table 1) corresponds with longer times in pools and higher exiting prices. These results indicating promising results for the model-implied liquidity provision strategy.

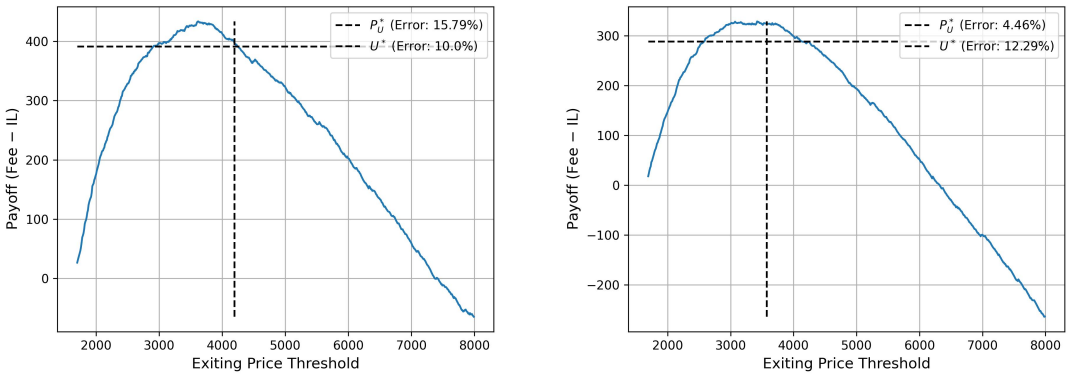

Figure 2: As an example, the $\bar{v}(P)$ functions (blue) and model-implied exit strategy results (black) are plotted for the WETH-USDT (left) and USDC-WETH (right) pool on starting at Aug. 1, 2022.

7 Conclusion

We study how the incentive of fee revenue and disincentive of impermanent loss affect liquidity provision using a optimal stopping approach. When using threshold policies to determine when to exit a liquidity pool, we find that LPs can recuperate impermanent loss by staying in the pool for downward price movements, but when facing upward price movements, there exists an upper threshold that maximizes expected payoffs relative to holding. At the optimum, the LP's expected payoff is positive, suggesting that before infrastructural fees, liquidity provision is profitable in expectation given that LPs exit liquidity pools strategically. The optimal upper price threshold can be used as a benchmark to evaluate pricing functions. Upon establishing a relationship between impermanent loss and the pricing function's curvature, we find that pricing functions with larger curvature given the same risky coin deposit amount correspond to higher optimal upper thresholds. When the fee renuve also depends on curvature, we provide conditions for which an unique optimal pricing function among a parametric family exists. These insights can be used by LPs for improved liquidity management and by DEXs for improved AMM design.

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A Technical Details

A.1 Pool Value Problem

Lemma A.1. *For any pricing function F that satisfies Assumption 1, the pool value problem under F at price P and invariant level $\mathcal{I}$ has a unique optimal solution.*

Proof. The pool value problem, reproduced below, is given by

$$V(P) = \min_{(x,y)\in\mathbb{R}^2_+} (Px + y)$$
$$\text{subject to } F(x,y) = \mathcal{I}.$$

The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = Px + y - \lambda(F(x,y) - \mathcal{I})$$

which has partial derivatives

$$\frac{\partial \mathcal{L}}{\partial x} = P - \lambda \cdot F_x(x,y)$$
$$\frac{\partial \mathcal{L}}{\partial y} = 1 - \lambda \cdot F_y(x,y)$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = \mathcal{I} - F(x,y)$$

Setting the partial derivatives to zero yields

$$\lambda \cdot F_x(x,y) = P$$
$$\lambda \cdot F_y(x,y) = 1$$
$$F(x,y) = \mathcal{I}$$

and dividing one equation by the other yields necessary conditions of

$$\frac{F_x(x,y)}{F_y(x,y)} = P$$
$$F(x,y) = \mathcal{I}.$$

We proceed to show that there exists a unique point on the curve $F(x,y) = \mathcal{I}$ that satisfies the equation $F_x(x,y)/F_y(x,y) = P$. Since $F_y > 0$ for all $(x,y) \in \mathbb{R}^2_+$, by the Implicit Function Theorem, the equation $F(x,y) - \mathcal{I} = 0$ parameterizes exists an implicit function $y(x)$ such that $F(x, y(x)) = \mathcal{I}$ for all $x \in \mathbb{R}_+$. Differentiating $F(x, y(x)) = \mathcal{I}$ with respect to x yields

$$F_x + F_y \cdot y'(x) = 0$$

$$y'(x) = -\frac{F_x}{F_y}\bigg|_{(x,y(x))}$$

Taking the derivative of F_x/F_y evaluated at $(x, y(x))$, we have

$$\frac{d}{dx}\left(\frac{F_x}{F_y}\right) = \frac{F_y(F_{xx} + F_{xy} \cdot y'(x)) - F_x(F_{xy} + F_{yy} \cdot y'(x))}{F_y^2}\bigg|_{(x,y(x))}.$$

Substituting the expression for $y'(x)$ yields

$$\frac{d}{dx}\left(\frac{F_x}{F_y}\right) = \frac{F_y^2 F_{xx} - 2F_x F_y F_{xy} + F_x^2 F_{yy}}{F_y^3}\bigg|_{(x,y(x))} < 0.$$

As $x \to 0$, by Assumption 1, $F_x/F_y \to \infty$ along the curve $F(x, y(x)) = \mathcal{I}$. Similarly, as $x \to \infty$, we have $F_x/F_y \to 0$ along the curve $F(x, y(x)) = \mathcal{I}$. Since F_x/F_y is decreasing in x along the curve $F(x, y(x)) = \mathcal{I}$ as shown above, there exists a unique point along the curve $F(x, y(x)) = \mathcal{I}$ such that $F_x/F_y = P$. Since there is a unique point satisfying the necessary condition, it follows that there exists a unique solution to the pool value problem. □

Lemma A.2. *Let $(x(P), y(P))$ and $V(P)$ be the optimal solution and value, respectively, to the pool value problem at price P and invariant level $\mathcal{I}$. Then*

- $V'(P) = x^*(P) > 0$
- $V''(P) = x^{*\prime}(P) \leq 0$

Proof. See Milionis et al. (2022). □

A.2 Motivation of Constant Fee Reward Rate

Lemma A.3. *Suppose that the combined trading intensity of the arbitrageurs and noise traders at time t is $\nu \cdot x^*(P_0, \mathcal{I}_t)$. Then the LPs fee reward intensity is constant.*

Proof. Suppose that the LP deposits (x_0, y_0) liquidity at time 0, and that time t, the LP's share of the pool is $1/(1 + w_t)$ where $w_t \in \mathbb{R}$ represents the entry/exit of other LPs in the pool. If the fundamental price were still P_0, then there would be $(1 + w_t)x_0$ risky coins and $(1 + w_t)y_0$ stable coins in the liquidity pool (by the 0-homogeneity of F_x/F_y), so the LP's fee reward intensity per unit of risky coin deposited is then

$$\begin{aligned}\frac{\text{Share of Pool} \times \text{Fee Tier} \times \text{Trade Intensity}}{\text{Risky Coins Deposited}} &= \frac{1}{x_0} \cdot \frac{1}{1 + w_t} \times f \times \nu \times x^*(P_0; \mathcal{I}_t) \\ &= \frac{1}{x_0} \cdot \frac{1}{1 + w_t} \times f \times \nu \times (1 + w_t)x_0 = f\nu := \phi\end{aligned}$$

which is constant, as desired. □

B Proofs

B.1 Proof of Theorem 4.1

We require the following lemma.

Lemma B.1. *If the CFMM's pricing function F satisfies Assumption 1, then*

- $\ell'(p) \geq \ell''(p)$ *for all* $p > 0$*;*
- $\lim_{p\to\infty}(\ell(p) - \ell'(p)) = \infty$.

Proof. We write the above conditions in terms of the absolute price given by the change-of-variables $P = P_0 e^p$. By the Chain Rule,

$$\ell'(p) = \frac{d\ell}{dp} = \frac{\partial L}{\partial P} \cdot \frac{\partial P}{\partial p} = L'(P) \cdot P_0 e^p = P \cdot L'(P)$$

$$\ell''(p) = \frac{d^2\ell}{dp^2} = \frac{d}{dp}(P \cdot \ell'(P)) = \frac{\partial}{\partial P}(P \cdot L'(P)) \cdot \frac{\partial P}{\partial p} = P^2 \cdot L''(P) + P \cdot L'(P)$$

The conditions can then be written as

- $L''(P) \leq 0$ for all $P > P_0$;
- $\lim_{P\to\infty}(L(P) - P \cdot L'(P)) = \infty$.

By Lemma A.3, without loss of generality, we can work with a representative LP that deposits x_0 risky coins and y_0 stable coins. We denote the optimal solution to the pool value problem at fundamental price P and invariant level $\mathcal{I} = F(x_0, y_0)$ as $(x(P), y(P))$. Then we can write the pool value and value from holding the same amount of assets, respectively, as

$$V(P) = P \cdot x(P) + y(P)$$
$$V_h(P) = P \cdot x(P_0) + y(P_0)$$

Recall that the per-unit impermanent loss is

$$L(P) = \frac{V(P) - V_h(P)}{x_0}$$

By Lemma A.1, we have

$$L'(P) = \frac{V'(P) - V_h'(P)}{x(P_0)} = \frac{x(P)}{x(P_0)} - 1$$
$$L''(P) = \frac{V''(P) - V_h''(P)}{x(P_0)} = \frac{x'(P)}{x(P_0)}$$

Since $x'(P) < 0$, it follows that $L''(P) \leq 0$.

As $y \to 0$, by Assumption 1, $F_x/F_y \to 0$ along the curve $F(x, y(x)) = \mathcal{I}$. Similarly, as $y \to \infty$, we have $F_x/F_y \to \infty$ along the curve $F(x, y(x)) = \mathcal{I}$. Since F_x/F_y is decreasing in x along the curve $F(x, y(x)) = \mathcal{I}$, it follows that F_x/F_y is increasing in y along the curve $F(x, y(x)) = \mathcal{I}$. Because we must have $F_x(x(P), y(P))/F_y(x(P), y(P)) = P$, it follows that $\lim_{P\to\infty} y(P) = \infty$, so

$$\lim_{P\to\infty} (y(P) - y(P_0)) = \lim_{P\to\infty} (L(P) - P \cdot L'(P)) = \infty$$

as desired. □

Let $\Lambda(\pi; p, p_L, p_U)$ be the discounted local time of $\{p_t\}$ at π given $p_0 = p$ and a stopping time of $T(p_L, p_U)$. That is, λ is the function such that

$$m(A; p, p_L, p_U) = \int_{p\in A} \Lambda(\pi; p, p_L, p_U)\, dp$$

where m is given by

$$m(A; p, p_L, p_U) = \int_0^{T(p_L,p_U)} e^{-rt} \mathbb{1}_A(p_t)\, dt.$$

Standard probability theory results (e.g. Chung and Williams (1990)) show that such a function Λ exists. Let $\psi_L(p; p_L, p_U)$ and $\psi_U(p; p_L, p_U)$ be the expected discount factors at $T(p_L, p_U)$ conditional on $p_{T(p_L,p_U)} = p_L$ and $p_{T(p_L,p_U)} = p_U$, respectively, and $p_0 = p$. That is,

$$\begin{aligned}
\psi_L(p; p_L, p_U) &= \mathbb{E}_p[e^{-rT(p_L,p_U)} | p_{T(p_L,p_U)} = p_L] \cdot \mathbb{P}_p(p_{T(p_L,p_U)} = p_L), \\
\psi_U(p; p_L, p_U) &= \mathbb{E}_p[e^{-rT(p_L,p_U)} | p_{T(p_L,p_U)} = p_U] \cdot \mathbb{P}_p(p_{T(p_L,p_U)} = p_U).
\end{aligned}$$

We can then write the LP's objective, denoted v, as

$$v(0; p_L, p_U) = \int_{p_L}^{p_U} \mathbb{E}[\Lambda(\pi; 0, p_L, p_U)]\, d\pi + \psi_L(0; p_L, p_U) \cdot \ell(p_L) + \psi_U(0; p_L, p_U) \cdot \ell(p_U).$$

For the remainder of the proof, we will use the following facts (see Stokey (2008)):

$$\begin{aligned}
\frac{\partial}{\partial p_L} \mathbb{E}[\Lambda(\pi; 0, p_L, p_U)] &= \psi_L(0; p_L, p_U) \cdot \frac{\partial}{\partial p_L} \mathbb{E}[\Lambda(\pi; p_L, p_L, p_U)], \\
\frac{\partial}{\partial p_U} \mathbb{E}[\Lambda(\pi; 0, p_L, p_U)] &= \psi_U(0; p_L, p_U) \cdot \frac{\partial}{\partial p_L} \mathbb{E}[\Lambda(\pi; p_U, p_L, p_U)], \\
\frac{\partial}{\partial p_L} \psi_L(0; p_L, p_U) &= \psi_L(0; p_L, p_U) \cdot \frac{\partial}{\partial p_L} \psi_L(p_L; p_L, p_U), \\
\frac{\partial}{\partial p_U} \psi_L(0; p_L, p_U) &= \psi_U(0; p_L, p_U) \cdot \frac{\partial}{\partial p_U} \psi_L(p_U; p_L, p_U), \\
\frac{\partial}{\partial p_L} \psi_U(0; p_L, p_U) &= \psi_L(0; p_L, p_U) \cdot \frac{\partial}{\partial p_L} \psi_U(p_L; p_L, p_U),
\end{aligned}$$

$$\frac{\partial}{\partial p_L}\psi_U(0;p_L,p_U) = \psi_U(0;p_L,p_U)\cdot\frac{\partial}{\partial p_U}\psi_U(p_U;p_L,p_U),$$

and the closed forms

$$\mathbb{E}[\Lambda(\pi;p,p_L,p_U)] = \begin{cases} \left(\frac{\sigma^2}{2}(\xi^+-\xi^-)\right)^{-1}\left(e^{-\xi^-(p-\pi)} - \psi_U e^{\xi^-(p_U-\pi)} - \psi_L e^{\xi^+(p_L-\pi)}\right) & \pi \le 0 \\ \left(\frac{\sigma^2}{2}(\xi^+-\xi^-)\right)^{-1}\left(e^{-\xi^+(p-\pi)} - \psi_U e^{\xi^-(p_U-\pi)} - \psi_L e^{\xi^+(p_L-\pi)}\right) & \pi \ge 0, \end{cases}$$

$$\psi_L(p;p_L,p_U) = \frac{e^{\xi^-p+\xi^+p_U} - e^{\xi^+p+\xi^-p_U}}{e^{\xi^-p_L+\xi^+p_U} - e^{\xi^+p_L+\xi^-p_U}},$$

$$\psi_U(p;p_L,p_U) = \frac{e^{\xi^-p+\xi^+p_U} - e^{\xi^+p+\xi^-p_U}}{e^{\xi^-p_L+\xi^+p_U} - e^{\xi^+p_L+\xi^-p_U}}.$$

Then the derivative of v with respect to p_L can be written as

$$\begin{aligned}\frac{\partial v(0)}{\partial p_L} = A\cdot((\xi^+e^{\xi^+p_L+\xi^-p_U} - \xi^-e^{\xi^+p_U+\xi^-p_L})(\ell(p_L)-\phi/r) + (\xi^+-\xi^-)(\ell(p_U)-\phi/r) \\ + \ell'(p_U)(e^{\xi^+p_U+\xi^-p_L} - e^{\xi^+p_L+\xi^-p_U}))\end{aligned}$$

where A is given by

$$A = \frac{e^{\xi^+p_U} - e^{\xi^-p_U}}{(e^{\xi^+p_U+\xi^-p_L} - e^{\xi^+p_L+\xi^-p_U})^2}.$$

Since $\xi^+, p_U > 0$ and $\xi^-, p_L < 0$, it follows that $\partial v(0)/\partial p_L < 0$. Then $p_L^\star = -\infty$.

Since the optimal choice of p_L is $-\infty$ for any $p_U > 0$, the LP's objective can be rewritten as

$$v(p_U) = \int_{-\infty}^{p_U}\mathbb{E}[\Lambda(\pi;0,-\infty,p_U)]\,d\pi + \psi_L(0;-\infty,p_U)\cdot\ell(-\infty) + \psi_U(0;-\infty,p_U)\cdot\ell(p_U).$$

The derivative of v with respect to p_U can be written as

$$\frac{\partial v(0)}{\partial p_U} = e^{-\xi^+p_U}\left(\frac{\phi}{r}\cdot\xi^+ - \xi^+\cdot\ell(p_U) + \ell'(p_U)\right),$$

so the first-order condition for this problem is

$$\mathsf{FOC}(p_U) = \frac{\phi}{r}\xi^+ - \xi^+\cdot\ell(p_U) + \ell'(p_U) = 0.$$

If $r \le \mu$, then $\xi^+ \le 1$. If $\xi^+ \le 1$, then for any $p_U > 0$,

$$0 > -\ell'(p_U) + \ell''(p_U) \ge -\xi^+\cdot\ell'(p_U) + \ell''(p_U) = \frac{\partial\mathsf{FOC}}{\partial p_U}(p_U),$$

so $\mathsf{FOC}(p_U)$ is decreasing in p_U by Lemma B.1. We also have

$$\frac{\phi}{r}\xi^+ - \ell(p_U) + \ell'(p_U) \geq \frac{\phi}{r}\xi^+ - \xi^+ \cdot \ell(p_U) + \ell'(p_U) = \mathsf{FOC}(p_U),$$

so $\lim_{p_U \to \infty} \mathsf{FOC}(p_U) = -\infty$ by Lemma B.1.

Since $\mathsf{FOC}(0) > 0$, $\mathsf{FOC}(p_U)$ is decreasing in p_U, and $\lim_{p_U \to \infty} \mathsf{FOC}(p_U) = -\infty$, there exists a unique $p_U^* \in (0, \infty)$ such that $\mathsf{FOC}(p_U^*) = 0$. Then $\mathsf{FOC}(p_U) > 0$ for all $p_U < p_U^*$ and $\mathsf{FOC}(p_U) < 0$ for all $p_U > p_U^*$, so

$$\frac{\partial v(0)}{\partial p_U} \begin{cases} > 0 & p_U < p_U^* \\ < 0 & p_U > p_U^* \end{cases}.$$

Since $v(0) = 0$ for $p_U = 0$ and $v(0)$ is increasing in p_U to p_U^*, it follows that $v(0; p_L^*, p_U^*) > 0$ for $p_U = p_U^*$ is positive.

We finally show that when $r > \mu$, then $p_U^* = \infty$. Note that $r > \mu$ if and only if $\xi^+ - 1$. We first show that $y'(P) = o(1)$. Suppose, towards a contradiction, that $|x'(P)| = \Omega(P^{-1})$. Then $|x(P)| = \Omega(\log P)$. Since $x(P)$ is positive and decreasing, we have a contradiction. Then we must have $|x'(P)| = o(P^{-1})$, so $|P \cdot x'(P)| = o(1)$. Since $y'(P) = -P \cdot x'(P)$, it follows that $|y'(P)| = o(1)$. Since $y(P)$ is increasing, the claim follows. Then

$$\begin{aligned} |L(P)| &= |P \cdot (x(P) - x(P_0)) + (y(P) - y(P_0))| \\ &\leq P \cdot |x(P) - x(P_0)| + |y(P) - y(P_0)| \leq P \cdot x(P_0) + |y(P) - y(P_0)|. \end{aligned}$$

Since $y'(P) = o(1)$, it follows that $|L(P)| = O(P)$.

Recall that

$$v(0; p_U) = \frac{\phi}{r} + e^{-\xi^+ p_U} \left(\ell(p_U) - \frac{\phi}{r} \right).$$

Changing variables from log-prices to absolute prices yields

$$v(0; p_U) = \frac{\phi}{r} + \left(\frac{P_U}{P_0} \right)^{-\xi^+} \left(L(P_U) - \frac{\phi}{r} \right).$$

Since $\xi^+ > 0$, we have

$$V(0; p_U) = O(P_U^{-\xi^+} \cdot L(P_U)).$$

It follows that if $\xi^+ > 1$, then $|P_U^{-\xi^+} \cdot L(P_U)| \to 0$ as $P_U \to \infty$. Then $V(0; p_U) \to \frac{\phi}{r}$ as $p_U \to \infty$, the maximal attainable value, so the optimal upper threshold when $\xi^+ > 1$ is $p_U^\star = \infty$.

B.2 Proof of Theorem 4.2

We first show that p_U^* is increasing in ϕ. The FOC is

$$\mathsf{FOC}(p_U) = \frac{\phi}{r}\xi^+ - \xi^+ \cdot \ell(p_U) + \ell'(p_U) = 0.$$

Recall in the proof of Theorem 4.1 that FOC is decreasing in p_U. Then by implicit differentiation,

$$\frac{dp_U^*}{d\phi} = -\frac{\frac{\xi^+}{r}}{\frac{\partial \mathsf{FOC}}{\partial p_U}(p_U^*)} > 0.$$

We then look at the comparative statics of p_U^* with respect to μ and σ. Recall that

$$\xi^+ = \frac{-\tilde{\mu} + \sqrt{\tilde{\mu}^2 + 2\sigma^2 r}}{2\sigma^2} = \frac{-(\mu - \sigma^2/2) + \sqrt{(\mu - \sigma^2/2)^2 + 2\sigma^2 r}}{2\sigma^2}.$$

Differentiating with respect to $\tilde{\mu}$ yields

$$\frac{\partial \xi^+}{\partial \tilde{\mu}} = \frac{1}{2\sigma^2}\left(\frac{\tilde{\mu}}{\sqrt{\tilde{\mu}^2 + 2r\sigma^2}} - 1\right) < 0$$

and differentiating with respect to σ yields

$$\frac{\partial \xi^+}{\partial \sigma} = -\frac{\mu\sqrt{(2\mu - \sigma^2)^2 + 8r\sigma^2} + (2r - \mu)\,\sigma^2 + 2\mu^2}{\sigma^3\sqrt{(2\mu - \sigma^2)^2 + 8r\sigma^2}} < 0.$$

By implicit differentiation,

$$\frac{dp_U^*}{d\mu} = -\frac{\left(\frac{\phi}{r} - \ell(p_U^*)\right) \cdot \frac{\partial \xi^+}{\partial \tilde{\mu}} \cdot \frac{\partial \tilde{\mu}}{\partial \mu}}{\frac{\partial \mathsf{FOC}}{\partial p_U}(p_U^*)} < 0,$$

$$\frac{dp_U^*}{d\sigma} = -\frac{\left(\frac{\phi}{r} - \ell(p_U^*)\right) \cdot \frac{\partial \xi^+}{\partial \sigma}}{\frac{\partial \mathsf{FOC}}{\partial p_U}(p_U^*)} < 0$$

as desired.

B.3 Proof of Proposition 5.1

For any invariant level $\mathcal{I}$, the arbitrage-free quantities $x(P)$ and $y(P)$ of the risky and stable coins, respectively, must satisfy

$$F(x(P), y(P)) - \mathcal{I} = 0$$
$$\frac{F_x(x(P), y(P))}{F_y(x(P), y(P))} - P = 0.$$

Taking derivatives with respect to P implies that $x'(P)$ and $y'(P)$ must satisfy

$$F_x \cdot x'(P) + F_y \cdot y'(P) = 0$$
$$\frac{F_y F_{xx} - F_x F_{xy}}{F_y^2} \cdot x'(P) + \frac{F_y F_{xy} - F_x F_{yy}}{F_y^2} \cdot y'(P) - 1 = 0.$$

Solving the first equation for $y'(P)$ yields

$$y'(P) = -\frac{F_x}{F_y} \cdot x'(P)$$
$$y(P) = y(P_0) + \int_{P_0}^{P} y'(\pi)\, d\pi$$

Plugging this expression into the second equation and solving for $x'(P)$ yields

$$x'(P) = \frac{F_y^3}{F_y^2 F_{xx} - 2F_x F_y F_{xy} + F_x^2 F_{yy}}.$$

Assume that $F_x, F_y, F_{xy} > 0$ and $F_{xx}, F_{yy} \leq 0$. Note that we can write

$$x'(P) = \frac{F_y^3}{F_y^2 F_{xx} - 2F_x F_y F_{xy} + F_x^2 F_{yy}} = -\frac{(F_x^2 + F_y^2)^{3/2}}{|F_y^2 F_{xx} - 2F_x F_y F_{xy} + F_x^2 F_{yy}|} \left(\frac{F_x^2 + F_y^2}{F_y^2} \right)^{-3/2}.$$

Let $\kappa(P)$ be the curvature of the implicit curve $F(x, y) - \mathcal{I} = 0$ at $(x(P), y(P))$. Then

$$x'(P) = -\frac{1}{\kappa(P)} \left(\frac{1}{1 + P^2} \right)^{3/2}$$
$$x(P) = x(P_0) + \int_{P_0}^{P} x'(\pi)\, d\pi.$$

Since F_x/F_y evaluated at $(x(P), y(P))$ yields the spot price P, plugging in the expression for $x'(P)$ into the expression for $y'(P)$ yields

$$y'(P) = \frac{P}{\kappa(P)} \left(\frac{1}{1 + P^2} \right)^{3/2}$$
$$y(P) = y(P_0) + \int_{P_0}^{P} y'(\pi)\, d\pi.$$

Since the per-unit impermanent loss is given by

$$L(P) = \frac{V(P) - V_h(P)}{x_0} = \frac{P(x(P) - x(P_0)) + (y(P) - y(P_0))}{x_0}$$

the result follows.

B.4 Proof of Theorem 5.2

Lower Curvature Implies Higher Upside Loss Since the curvature of a nonlinear implicit curve is positive, $x'(P) < 0$ and $y'(P) > 0$. Since

$$L'(P) = \frac{1}{x(P_0)}(x(P) - x(P_0)) = \frac{\int_{P_0}^{P} x'(\pi)\, d\pi}{x(P_0)},$$

for (nonlinear) pricing functions $F^{(1)}$ and $F^{(2)}$, if the LP deposits the same amount $x(P_0)$ of risky coins and the curvature of $F^{(1)}$ at implied invariant level $\mathcal{I}^{(1)}$ is greater than that of $F^{(2)}$ at implied invariant level $\mathcal{I}^{(2)}$ for every price $P > 0$, then the per-unit impermanent loss under $F^{(2)}$ is greater than that under $F^{(1)}$ for all $P > P_0$.

Higher Curvature Implies Higher Upper Threshold Recall that the first-order condition for p_U is

$$\mathsf{FOC}(p_U) = \frac{\phi}{r}\xi^+ - \xi^+ \cdot L(p_U) + L'(p_U) = 0.$$

Fix P_0 and x_0. Let $F^{(1)}$ and $F^{(2)}$ be nonlinear pricing functions with induced impermanent loss functions L_1 and L_2 at invariant levels $\mathcal{I}_1$ and $\mathcal{I}_2$, respectively, such that $x_1(P_0) = x_2(P_0) = x_0$. Suppose that $F^{(2)}$ has higher curvature than $F^{(1)}$ in the upside region, i.e. $P \in [P_0, \infty)$. Since FOC is decreasing, an equivalent condition for $p_U^*(F^{(1)}) < p_U^*(F^{(2)})$ is

$$\begin{aligned} -\xi^+ \cdot \ell_1(p_U) + \ell_1'(p_U) &\leq -\xi^+ \cdot \ell_2(p_U) + \ell_2'(p_U) \\ \xi^+(\ell_2(p_U) - \ell_1(p_U)) &\leq \ell_2'(p_U) - \ell_1'(p_U). \end{aligned}$$

Since lower curvature implies higher upside loss, $\ell_2(p_U) - \ell_1(p_U) > 0$ for $p_U > 0$. Since $r \leq \mu$ implies $\xi^+ \leq 1$, a sufficient condition for the above is

$$\ell_2(p_U) - \ell_1(p_U) \leq \ell_2'(p_U) - \ell_1'(p_U).$$

Changing variables to absolute prices yields

$$\begin{aligned} L_2(P_U) - L_1(P_U) &\leq P_U(L_1'(P_U) - L_2'(P_U)) \\ L_2(P_U) - P_U \cdot L_2'(P_U) &\leq L_1(P_U) - P_U \cdot L_1'(P_U) \\ \frac{1}{x_2(P_0)} \cdot (y_2(P_U) - y_2(P_0)) &\leq \frac{1}{x_1(P_0)} \cdot (y_1(P_U) - y_1(P_0)) \\ \int_{P_0}^{P_U} y_2'(P)\, dP &\leq \int_{P_0}^{P_U} y_1'(P)\, dP \end{aligned}$$

assuming that $x_1(P_0) = x_2(P_0)$, i.e. the LP deposits the same amount of risky coins for both pricing functions. The last inequality holds by the formula for $y'(P)$.

B.5 Proof of Theorem 5.3

Recall that in absolute prices,

$$\begin{aligned} L(P;\theta) &= x_0^{-1}(P(x(P)-x(P_0))+(y(P)-y(P_0))) \\ &= x_0^{-1}\left(P\int_{P_0}^{P} x'(\pi;\theta)\,d\pi + \int_{P_0}^{P} y'(\pi,\theta)\,d\pi\right) \end{aligned}$$

and that

$$\begin{aligned} L'(P;\theta) &= x_0^{-1}(x(P)-x(P_0)) \\ &= x_0^{-1}\int_{P_0}^{P} x'(\pi;\theta)\,d\pi. \end{aligned}$$

Impermanent Loss Part of FOC is Increasing Note that

$$\frac{\partial}{\partial\theta}\ell(p;\theta) \le \frac{\partial}{\partial\theta}\ell'(p;\theta) \implies -\xi^+\cdot\frac{\partial}{\partial\theta}\ell(p;\theta)+\frac{\partial}{\partial\theta}\ell(p;\theta)\ge 0.$$

Changing to absolute price means that we want to show that

$$\frac{\partial}{\partial\theta}L(P;\theta) \le \frac{\partial}{\partial\theta}P\cdot L'(P;\theta).$$

Substituting expressions yields

$$x_0^{-1}\left(P\int_{P_0}^{P}\frac{\partial}{\partial\theta}x'(\pi;\theta)\,d\pi+\int_{P_0}^{P}\frac{\partial}{\partial\theta}y'(\pi;\theta)\,d\pi\right)\le x_0^{-1}P\int_{P_0}^{P}\frac{\partial}{\partial\theta}x'(\pi;\theta)\,d\pi.$$

Since $1/\kappa(P;\theta)$ is decreasing, substituting the formula for $y'(\pi;\theta)$ yields that the second term on the LHS is negative, making the inequality true.

Impermanent Loss Part of FOC is Concave Note that

$$\frac{\partial^2}{\partial\theta^2}\ell'(p;\theta) \le \frac{\partial^2}{\partial\theta^2}\ell(p;\theta) \implies -\xi^+\cdot\frac{\partial^2}{\partial\theta^2}\ell(p;\theta)+\frac{\partial^2}{\partial\theta^2}\ell(p;\theta)\le 0.$$

Changing to absolute prices means that we want to show that

$$\frac{\partial^2}{\partial\theta^2}P\cdot L'(P;\theta) \le \frac{\partial^2}{\partial\theta^2}L(P;\theta).$$

Substituting expressions yields

$$x_0^{-1}P\int_{P_0}^{P}\frac{\partial^2}{\partial\theta^2}x'(\pi;\theta)\,d\pi \le x_0^{-1}\left(P\int_{P_0}^{P}\frac{\partial^2}{\partial\theta^2}x'(\pi;\theta)\,d\pi+\int_{P_0}^{P}\frac{\partial^2}{\partial\theta^2}y'(\pi;\theta)\,d\pi\right).$$

Since $1/\kappa(P;\theta)$ is convex, substituting the formula for $y'(\pi;\theta)$ yields that the second term on the RHS is positive, making the inequality true.

Remainder of Proof Fix $p_U \in \mathbb{R}_+$. By the first-order condition, if θ satisfies

$$-\xi^+ \cdot \ell(p_U;\theta) + \ell'(p_U;\theta) = -\phi(\theta) \cdot \frac{\xi}{r}$$

then the optimal upper threshold when the pricing function selected from the family of pricing functions has parameter θ is p_U.

Since $\lim_{p_U \to \infty} -\xi^+ \cdot \ell(p_U;\theta) + \ell'(p_U;\theta) \searrow -\infty$ for all θ, there exists a unique $\underline{p}_U$ such that $-\xi^+ \cdot \ell(\underline{p}_U;\underline{\theta}) + \ell'(\underline{p}_U;\underline{\theta}) = -\phi(\underline{\theta})$. Since $-\xi^+ \cdot \ell(\underline{p}_U;\underline{\theta}) + \ell'(\underline{p}_U;\underline{\theta})$. Since $-\xi^+ \cdot \ell(\underline{p}_U;\underline{\theta}) + \ell'(\underline{p}_U;\underline{\theta})$ increases but remains negative, $-\phi(\theta)$ increases to zero, and a convex and concave line intersect at most twice, there exists a unique $\theta > \underline{\theta}$ such that $-\xi^+ \cdot \ell(\underline{p}_U;\theta^*) + \ell'(\underline{p}_U;\theta) = \phi(\theta)$. Thus, $\underline{p}_U$ is an optimal threshold for $\underline{\theta}$ and θ. Since $\ell(p;\theta)$ decreases pointwise in p for all θ, the first θ solution increases and the second θ solution decreases as p increases. Since $\ell(p;\theta)$ is continuous, there exists some p_U^* where both solutions coincide at a point denoted θ^*. After that, i.e. for $p_U > p_U^*$, there is no value of θ that supports p_U as an optimal threshold. Thus, p_U is optimized at θ^*.